

Backends in Dart

QConSF - Oct 2022 / QCon Plus - Nov 2022

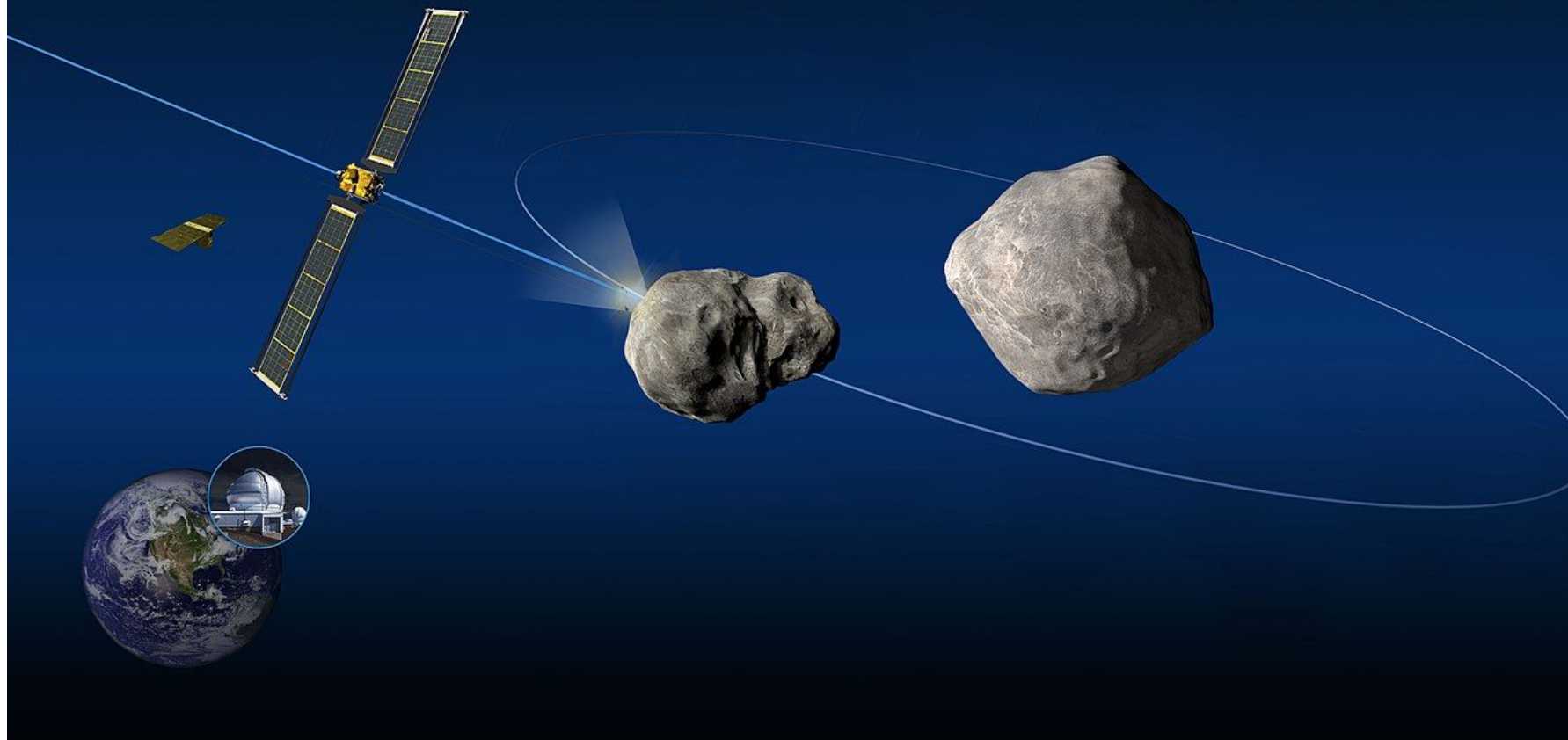






DART

Double Asteroid Redirection Test





Dart

“Dart is a client-optimized language
for fast apps on any platform”

@ dart.dev





**JIT
VM**





What I do

1. Improve the quality of documentation, samples and examples, so that atSigns are the easiest way for developers to build apps that have privacy preserving identity.
2. Build Continuous Delivery pipelines as a first step towards Progressive Delivery and Chaos Engineering so that we can confidently deploy high quality well tested software.
3. Lay the foundations for an engineering organisation that's optimised for high productivity by minimising cognitive load on teams and the people working in those teams.



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Agenda

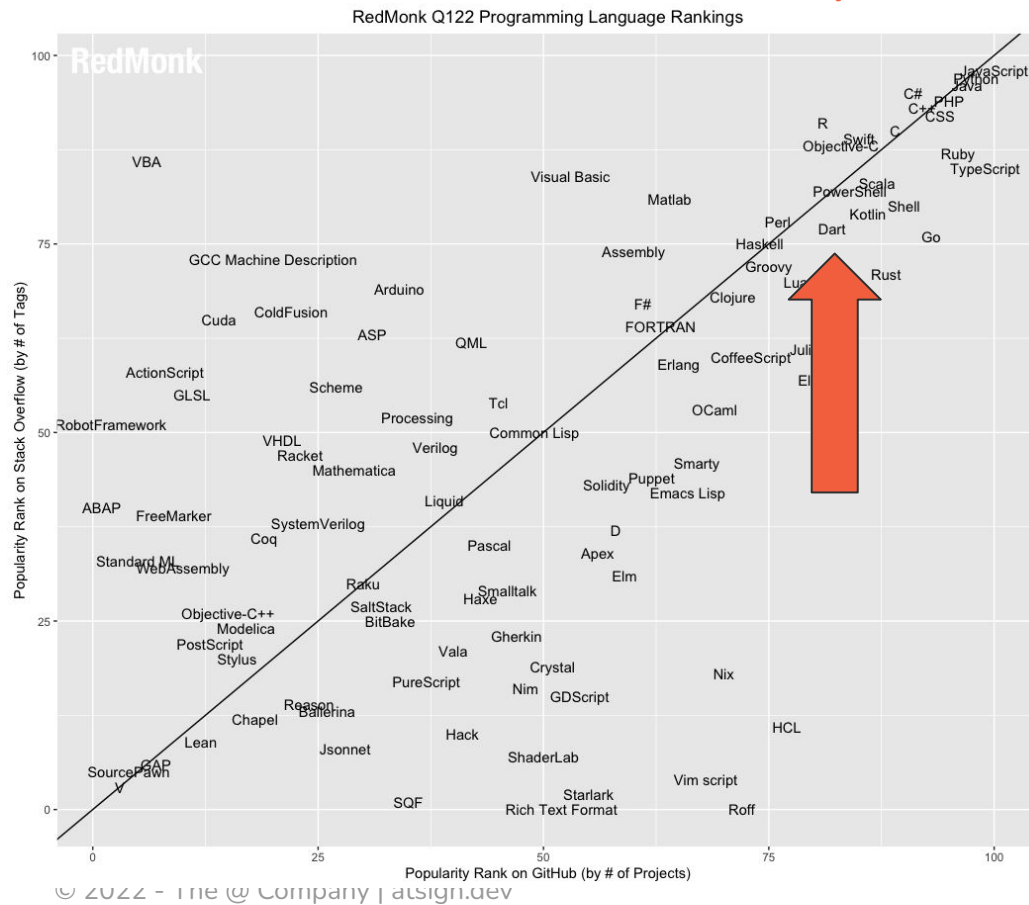
- Why Dart?
- Language features
- Just in time (JIT) vs Ahead of time (AOT)
- Dart in Containers
- Profiling and performance management
- A middle way with jit-snapshot?



Why Dart? Industry big picture.

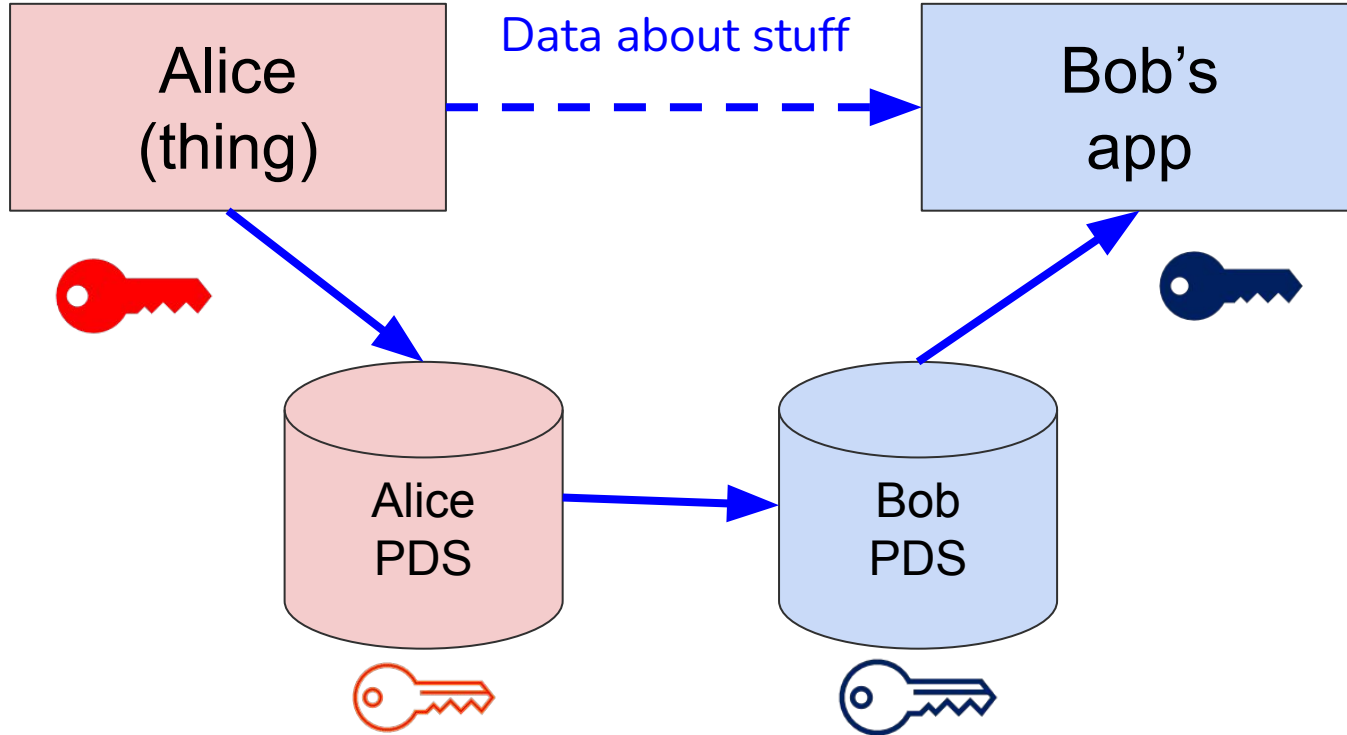
Flutter pulled Dart into the Redmonk Top 20

- 1 JavaScript
- 2 Python
- 3 Java
- 4 PHP
- 5 CSS
- 5 C#
- 7 C++
- 8 TypeScript
- 9 Ruby
- 10 C
- 11 Swift
- 12 R
- 13 Objective-C
- 14 Shell
- 14 Scala
- 16 Go
- 17 PowerShell
- 18 Kotlin
- 19 Rust
- 19 Dart



Why Dart for Atsign?

atSigns are personal data stores



Dart AOT binaries produce really small containers



atsigncompany/secondary ☆

By [atsigncompany](#) • Updated an hour ago

@protocol secondary server

Container

TAG

[dess](#)

`docker pull atsigncompany/secondary:...`



Last pushed 2 days ago by [atcpswan](#)

DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

[197123ec5e7a](#)

linux/amd64

5.97 MB

[1a75e98e541f](#)

linux/arm/v7

5 MB

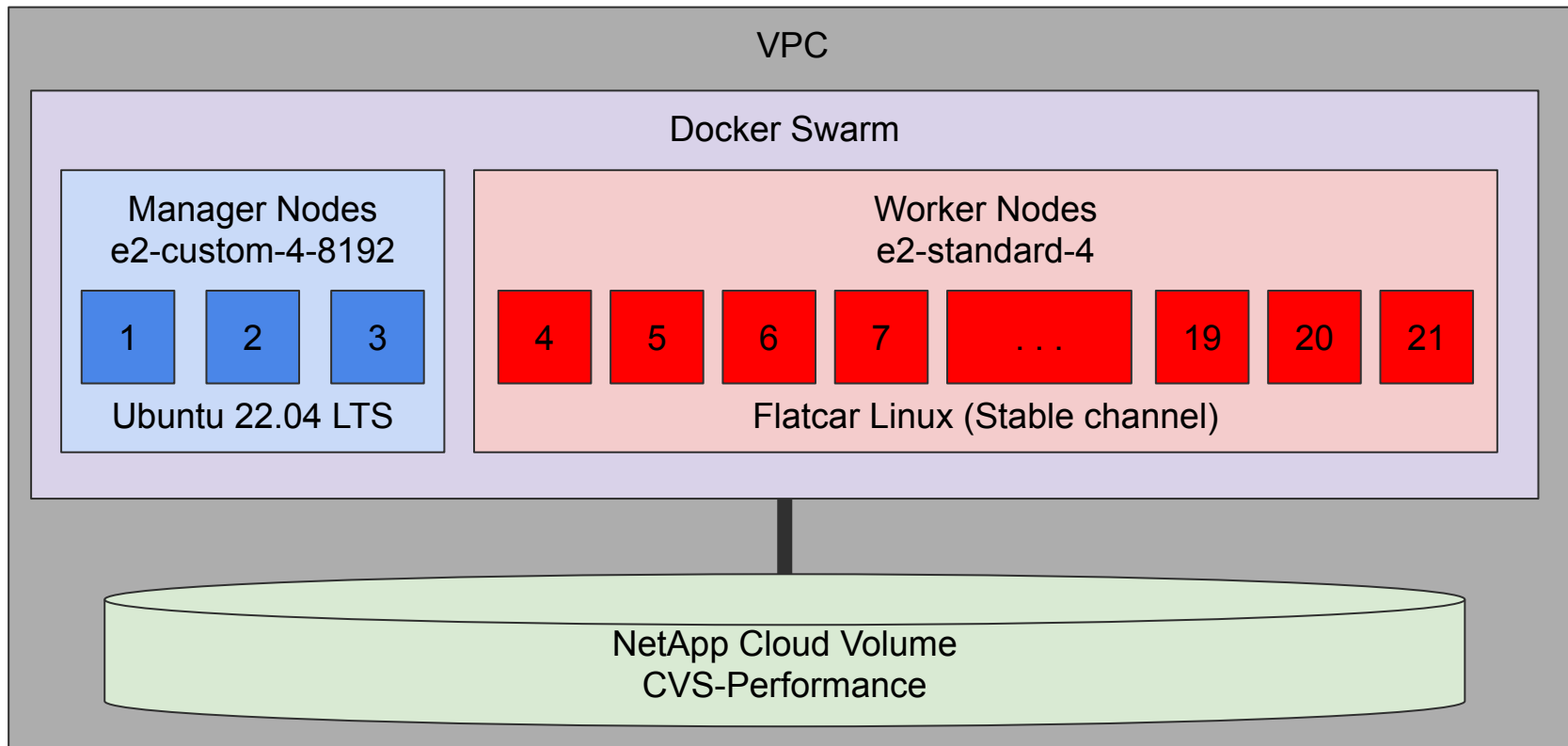
[6b20a465d6cd](#)

linux/arm64

5.21 MB



Infrastructure for ~6500 atSigns



Language features

'Hello World!'

```
void main() {  
    print('Hello, World!');  
}
```



Async/await (like JavaScript)

```
Future<String> createOrderMessage() async {  
    var order = await fetchUserOrder();  
    return 'Your order is: $order';  
}
```

```
Future<String> fetchUserOrder() =>  
    // Imagine that this function is  
    // more complex and slow.  
    Future.delayed(  
        const Duration(seconds: 2),  
        () => 'Large Latte',  
    );
```

```
Future<void> main() async {  
    print('Fetching user order...');  
    print(await createOrderMessage());  
}
```

Concurrency - Isolates

“independent workers that are similar to threads but don’t share memory, communicating only through messages”



Sound null safety (since Dart 2.12, Mar 2021)

```
// In null-safe Dart, none of these can ever be null.  
var i = 42; // Inferred to be an int.  
String name = getFileName();  
final b = Foo();  
  
// To indicate that a variable might have the value null  
// just add ? to its type declaration  
int? aNullableInt = null;
```



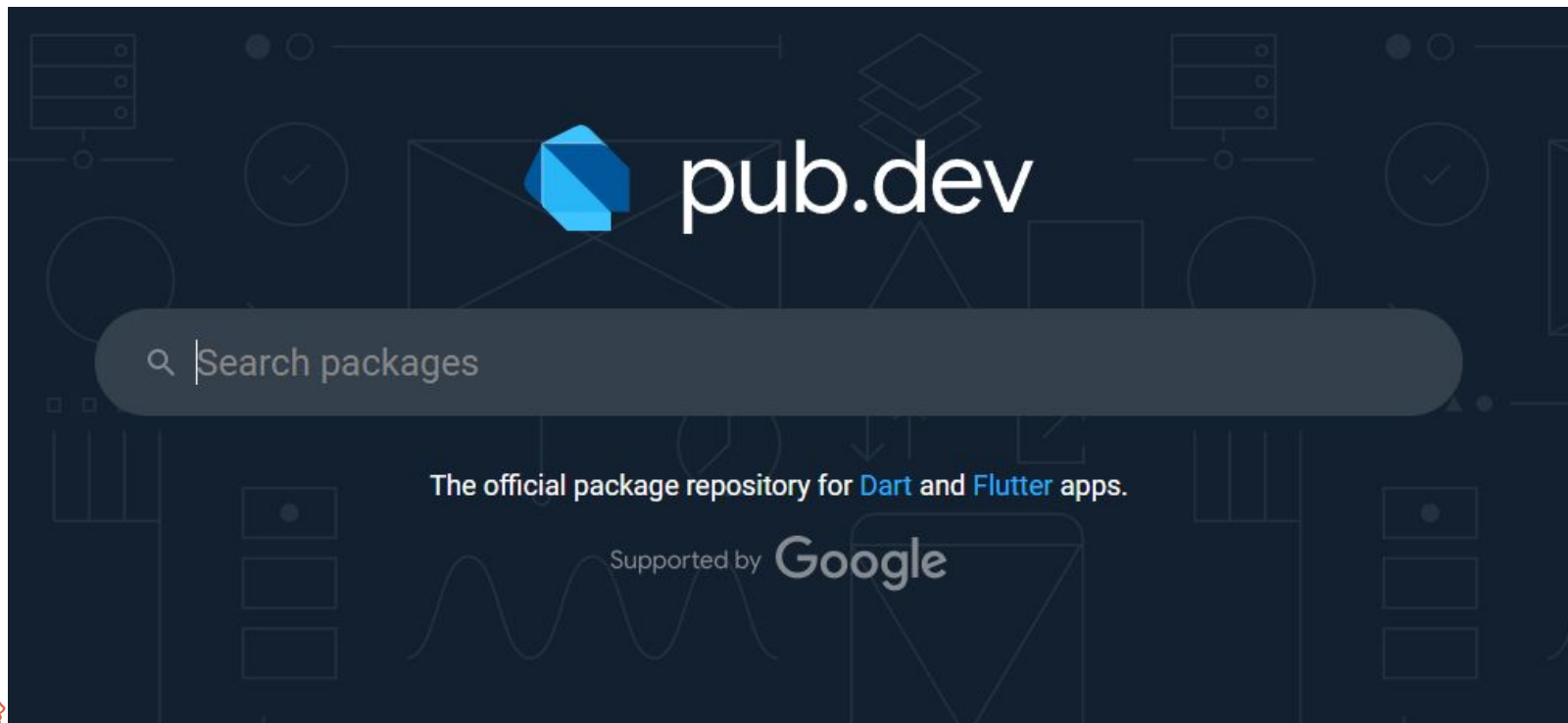
Licensing

BSD

license BSD3



Package Manager - pub.dev



JIT vs AOT

Dartshowplatform - A more useful 'Hello World!'

```
import 'dart:io' show Platform, stdout;

void main() {
  print(Platform.version);
}
```

```
2.18.2 (stable) (Tue Sep 27 13:24:11
2022 +0200) on "linux_arm64"
```



JIT - Just `dart run` it in the virtual machine

```
$ time dart run showplatform.dart
```

```
2.18.2 (stable) (Tue Sep 27 13:24:11  
2022 +0200) on "linux_arm64"
```

```
real    0m6.588s  
user    0m5.862s  
sys     0m0.925s
```



AOT - Compile it first then run the binary

```
$ dart compile exe showplatform.dart  
$ time ./showplatform.exe
```

```
2.18.2 (stable) (Tue Sep 27 13:24:11  
2022 +0200) on "linux_arm64"
```

```
real    0m0.027s  
user    0m0.012s  
sys     0m0.018s
```



Trade off - compilation is slow

```
$ time dart compile exe \  
showplatform.dart
```

```
Info: Compiling with sound null safety  
Generated: showplatform.exe
```

```
real      0m18.184s  
user      0m22.533s  
sys       0m2.643s
```



Dart in Containers

A typical Dockerfile for AOT compiled Dart binary

```
FROM dart AS build
WORKDIR /app
COPY ./mydemoapp.dart .
RUN dart pub get
RUN dart compile exe /app/mydemoapp.dart \
    -o /app/mydemoapp

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /app/mydemoapp /app/mydemoapp
ENTRYPOINT ["/app/mydemoapp"]
```



Image sizes for a trivial application



atsigncompany/dartshowplatform ☆

By [atsigncompany](#) • Updated 6 days ago

Multi arch Dart test image that prints out version and platform info

Container

TAG

latest

`docker pull atsigncompany/dartshowpl...`



Last pushed 6 days ago by [atcpswan](#)

DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

[8f4d6b9d1d68](#)

linux/amd64

4.45 MB

[100829c90d10](#)

linux/arm/v7

3.26 MB

[1dac2840cb6e](#)

linux/arm64

3.68 MB



Image sizes for a non trivial application



atsigncompany/secondary ☆

By [atsigncompany](#) • Updated an hour ago

@protocol secondary server

Container

TAG

[dess](#)

`docker pull atsigncompany/secondary:...`



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DIGEST

OS/ARCH

COMPRESSED SIZE ⓘ

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5.97 MB

[1a75e98e541f](#)

linux/arm/v7

5 MB

[6b20a465d6cd](#)

linux/arm64

5.21 MB



Profiling and performance management

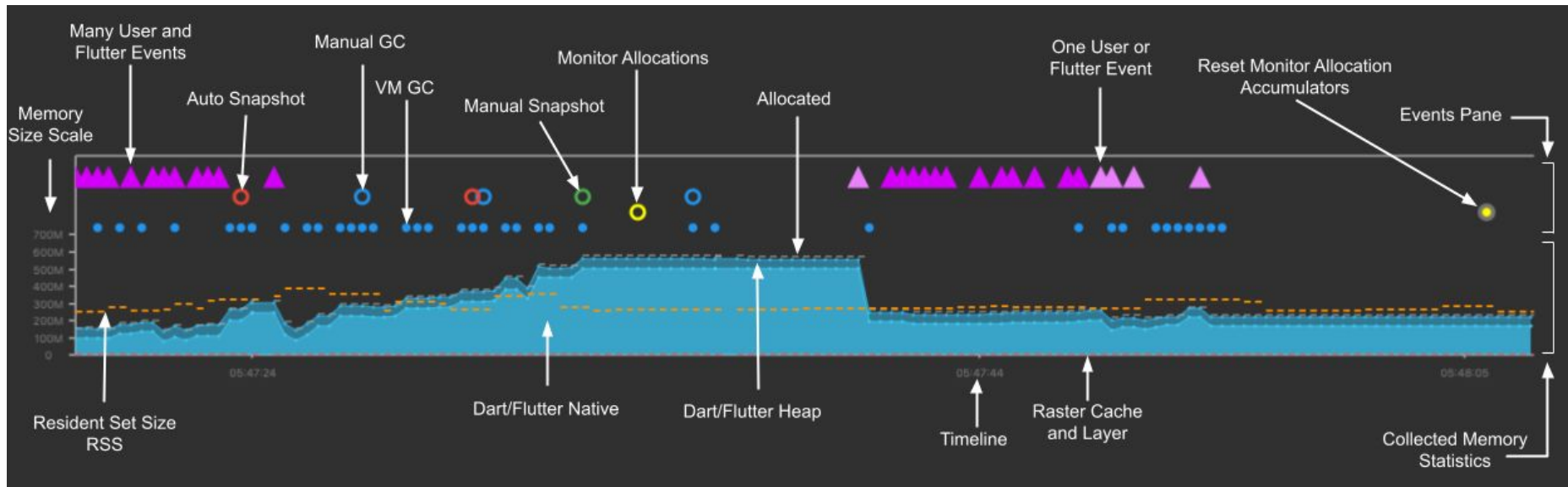
Dart DevTools

	Flutter mobile or desktop ↗	Flutter web ↗	Other web	Command-line
Debugger ↗	✓	✓	✓	✓
Logging view ↗	✓	✓	✓	✓
App size tool ↗	✓			✓
CPU profiler ↗	✓			✓
Memory view ↗	✓			✓
Network view ↗	✓			✓
Performance view ↗	✓			✓
Flutter inspector ↗	✓	✓		



<https://dart.dev/tools/dart-devtools>

Memory view



<https://flutter.dev/docs/development/tools/devtools/memory>



But... DevTools need to connect to a VM, so JIT only

[@mraleph](#) where would you start with trying to track down a potential memory leak in an AOT Dart app (running in a container)?

I'm presently planning to run an instance of it in a JIT VM so I can use the memory profiler, but if there's a way to get metrics from an AOT binary?

6:20 pm · 12 Aug 2021 · Twitter Web App



Vyacheslav Egorov @mraleph · 12 Aug



Replying to [@cpswan](#)

yeah, this is a bit complicated - if it was a Flutter application you could use so called profile mode (AOT with profiling tools enable) to collect a heap profile. but dart compile exe always produces release mode snapshots.



1



1



Vyacheslav Egorov @mraleph · 12 Aug



if you feel adventurous you could build your own SDK and change dart compile to produce vmervice enabled binaries.



A middle way?
jit-snapshot

Creating and running a jit-snapshot

```
$ dart compile jit-snapshot showplatform.dart  
$ time dart run showplatform.jit
```

```
2.18.2 (stable) (Tue Sep 27 13:24:11  
2022 +0200) on "linux_arm64"
```

```
real    0m1.786s  
user    0m1.783s  
sys     0m0.341s
```



Putting a jit-snapshot into a container

```
FROM dart AS build
WORKDIR /app
COPY ./mydemoapp.dart .
RUN dart pub get
RUN dart compile jit-snapshot /app/mydemoapp.dart \
    -o /app/mydemoapp.jit

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /usr/lib/dart/bin/ /usr/lib/dart/bin/
COPY --from=build /app/mydemoapp.jit /app/mydemoapp.jit
ENTRYPOINT ["/usr/lib/dart/bin/dart", "run",
"--observe=8181/0.0.0.0", "/app/mydemoapp.jit"]
```



Container image sizes

REPOSITORY	TAG	SIZE
atsigncompany/secondary	dev_obs	1.25GB
atsigncompany/secondary	prod	14.5MB
cpswan/secondary	obsjit	430MB



Runtime resource usage

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O
f938de4395a7	ss secondary.1.7wnwoifieezhdc91gkdwqnn12	0.04%	8.359MiB / 976.9MiB	0.86%	2.56kB / 5.97kB
6043df1c0027	cpsbh secondary.1.xkfk5w2gl76vhi8ifx8t94z37	0.05%	87.47MiB / 976.9MiB	8.95%	3.03kB / 6.05kB





Review

- Why Dart?
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Call to action: Try Dart yourself

Codelabs



The Dart codelabs provide a guided, hands-on coding experience—no download required!

General

Intro to Dart for Java Developers

Use DartPad to explore how Dart makes writing apps easy and fun.



Dart cheatsheet

Use DartPad to learn or remind yourself of some of the most commonly used, yet unique features of the Dart language.



<https://dart.dev/codelabs>



Thanks for your time

chris@atsign.com



@cpswan



Questions?